

CHARACTERIZATION OF ELASTOPLASTIC PROPERTIES OF ADDITIVELY MANUFACTURED SPECIMENS FROM INDENTATION DATA USING STOCHASTIC INVERSE MODELING

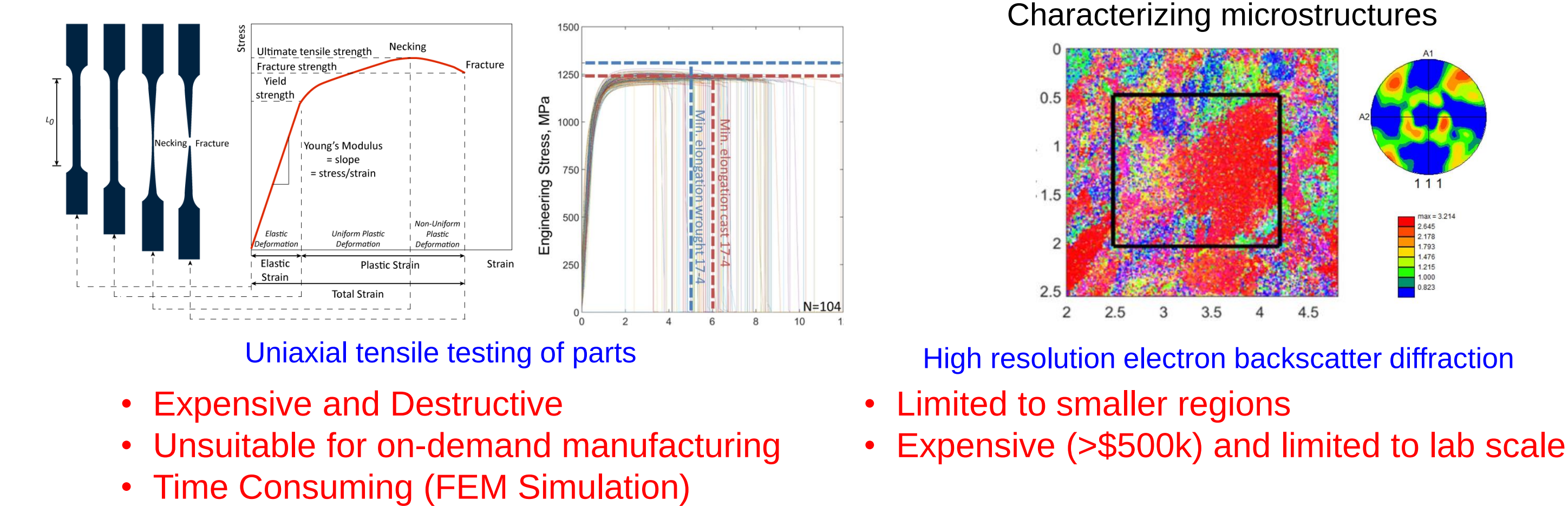
CHALLENGES WITH AM CHARACTERIZATION

- Near-net shape manufacturing from design; Efficient and scalable production of parts
- High-cost characterization



CONVENTIONAL C&Q IN ADDITIVE MANUFACTURING

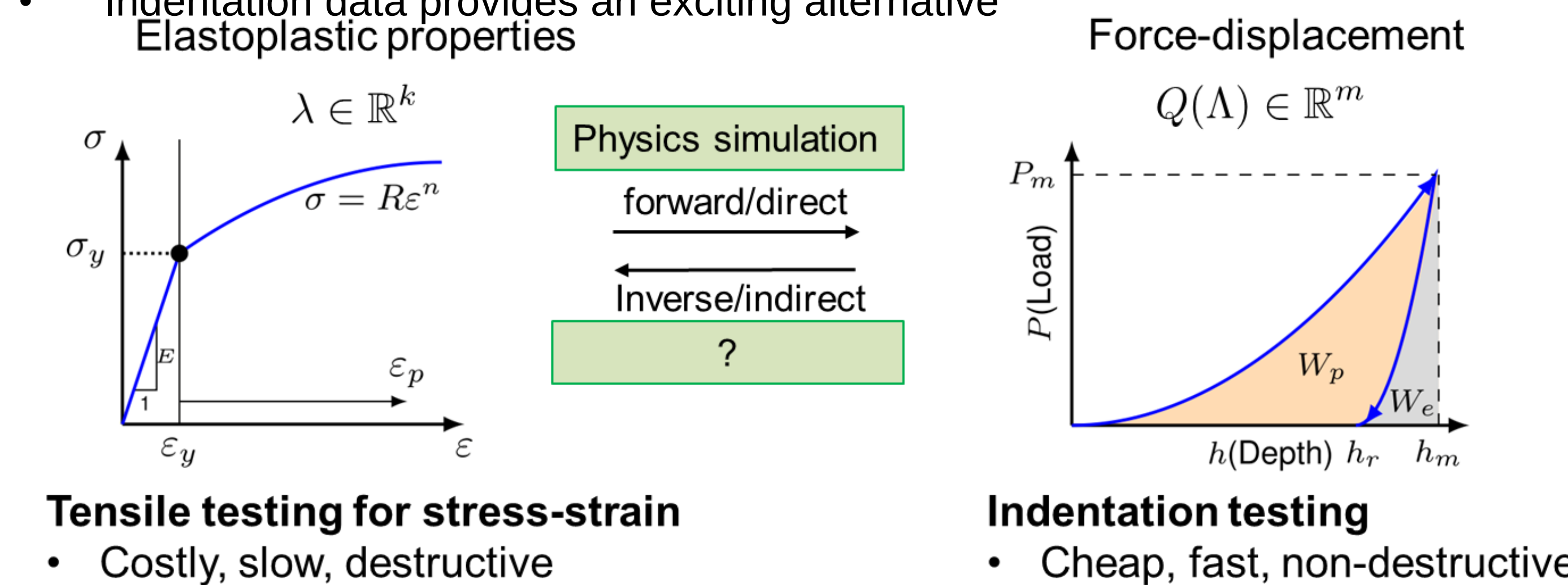
- Estimate thermal and mechanical properties (e.g., yield stress); Ensure part integrity
- Traditional C&Q is time-consuming and destructive
- Assessing tensile properties of manufactured parts



Is there an alternative approach to accelerate characterization?

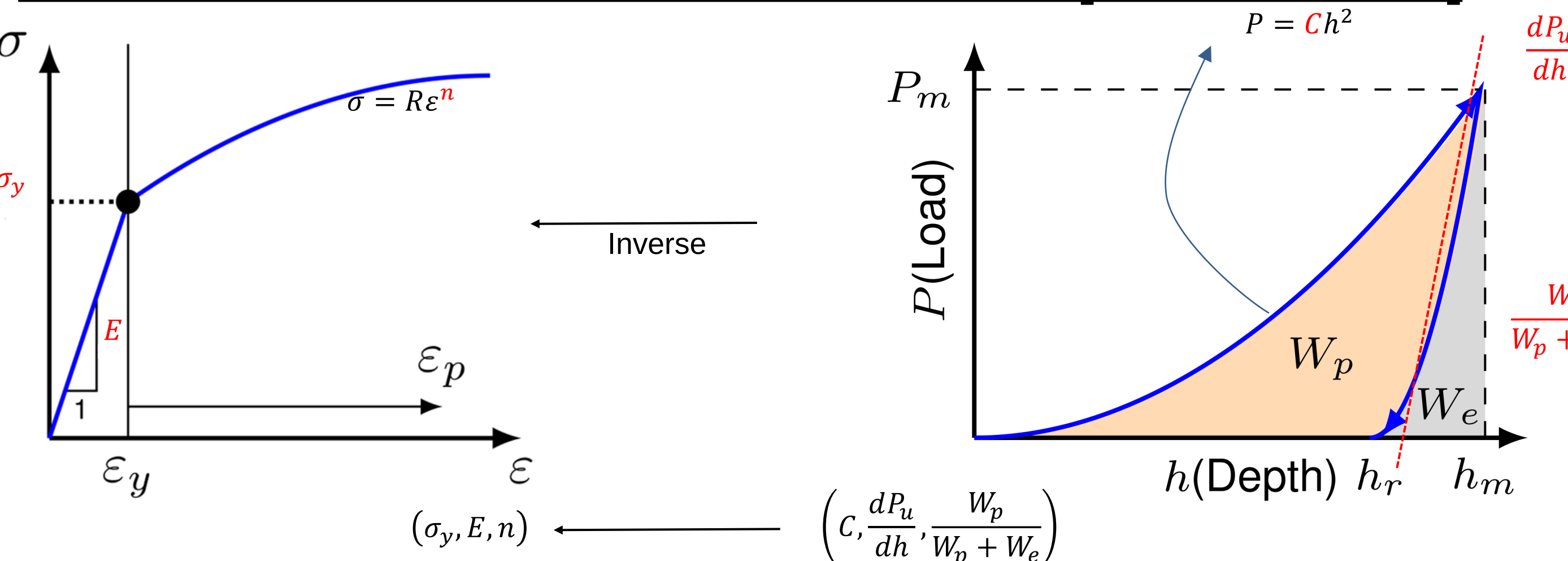
ALTERNATIVE APPROACH TO CHARACTERIZATION

- Elastoplastic properties is derived from stress-strain curve by uniaxial tensile test
- Indentation data provides an exciting alternative Elastoplastic properties



Can we indirectly estimate the properties?

EMPIRICAL APPROACH: FUNCTION-FITTING METHOD [DAO ET AL. 2001]



We need a generalizable approach that captures uncertainties

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ABSTRACT:

Rapid characterization of mechanical properties and material structure of additively manufactured (AM) components via non-destructive techniques (NDT) is essential for their wider adoption but remains challenging. Our work focuses on characterizing elastoplastic properties of AM components using instrumented indentation, known as the inverse indentation problem. Existing models often generalize poorly or lack a framework for consistent variability estimation. We propose the stochastic inverse problem (SIP), estimating a distribution of elastoplastic properties that matches observed indentation data. Applied to AM components with different heat treatments, our methodology predicts the average strain hardening exponent, Young's modulus, and yield strength within 1.1%, 1%, and 5% of actual values. Additionally, recovered property distributions closely match standard tensile tests, indicated by near-zero Kullback-Leibler divergence values.

BAYESIAN APPROACH

- Aims to estimate the uncertainty in the input (tensile properties) due to noisy measurements

$$\hat{\pi}_{\Lambda}^{\text{post}}(\lambda|q) \propto \pi_{\Lambda}^{\text{prior}}(\lambda)\pi(q|\lambda)$$

Limitation:

- Assumes that the variability in the data is due to measurement noise

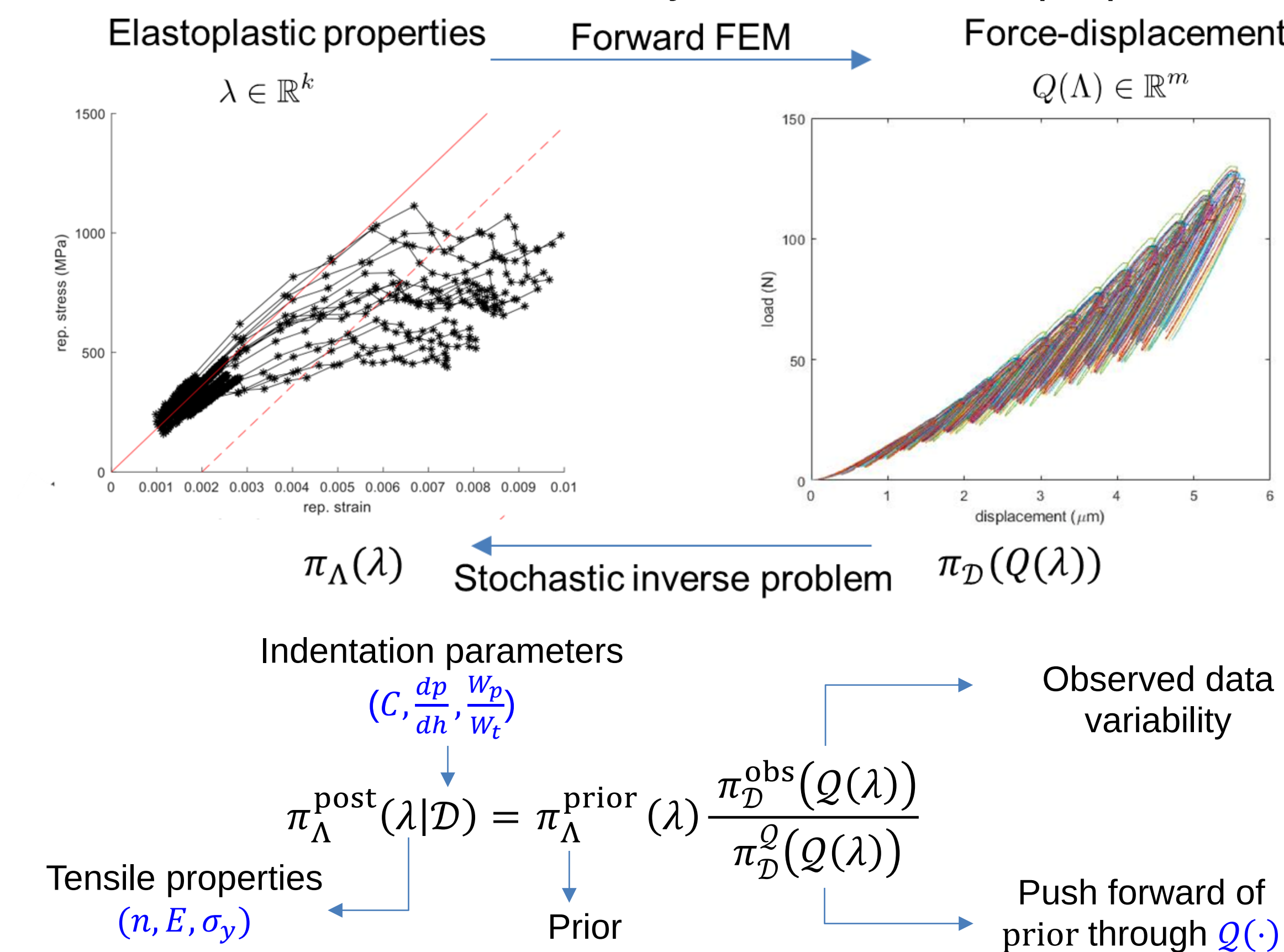
Consequence:

- More data will reduce uncertainty band leading to a point estimate

This approach is unable to recover true variability

METHOD: STOCHASTIC INVERSE MODELING

Objective: Estimate the variability of the unknown properties



Estimating distribution: rejection sampling $r(Q(\lambda_k)) = \frac{\pi_D^{\text{obs}}(Q(\lambda_k))}{\pi_D^Q(Q(\lambda_k))}$

- Generate samples from the prior distribution
- Compute the push-forward density induced by the prior
- Accept samples from the posterior distribution

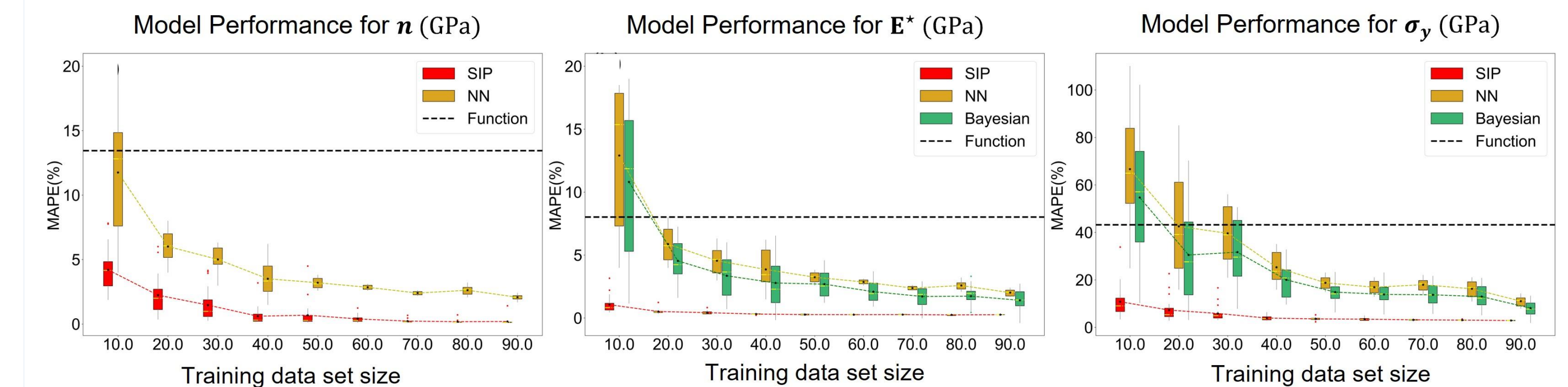
EXPERIMENTAL STUDY: AM STAINLESS STEEL WITH VARYING HEAT TREATMENT

Group Nomenclature:

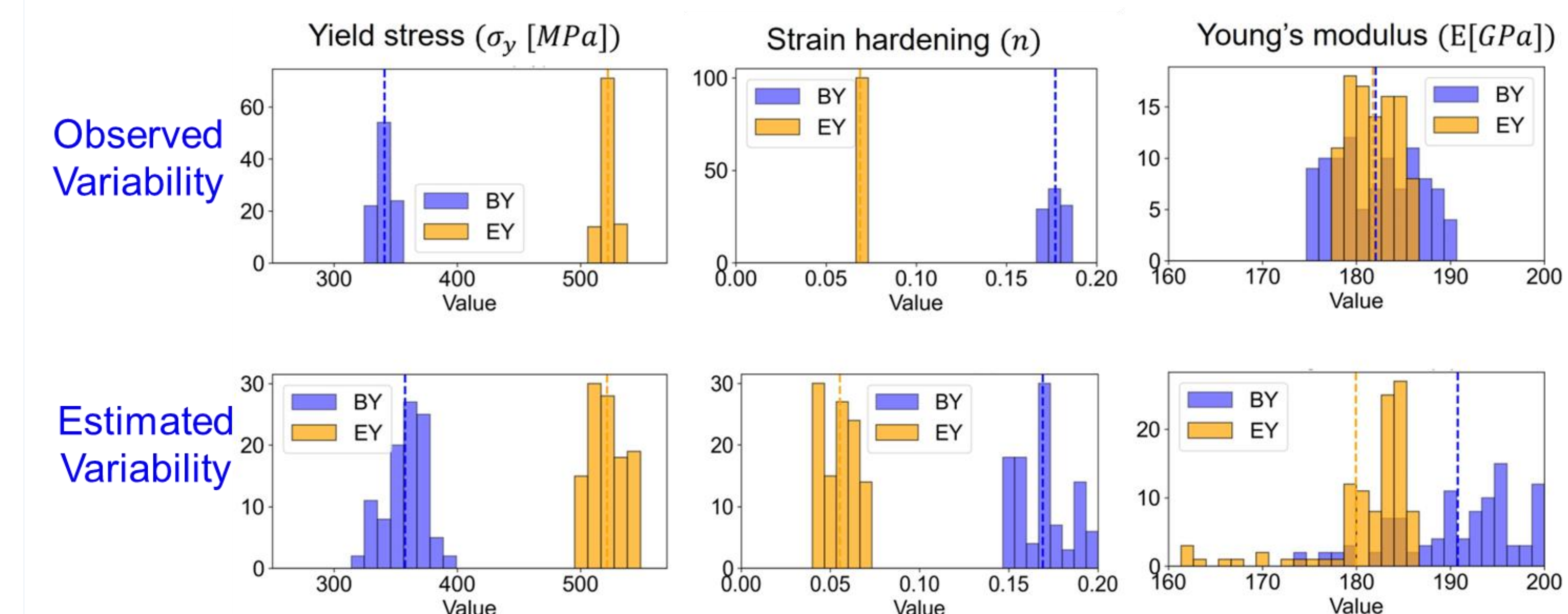
- **BY**: solution annealed at 1121° C for 2 hours followed by water quenching
- **EY**: stress-relieved at 650° C for 1 hour and then quenched in air

Objectives:

- Estimate the properties of the AM components and their variability
- Distinguish between properties of the two groups



MAPE values of different methods for different tensile properties

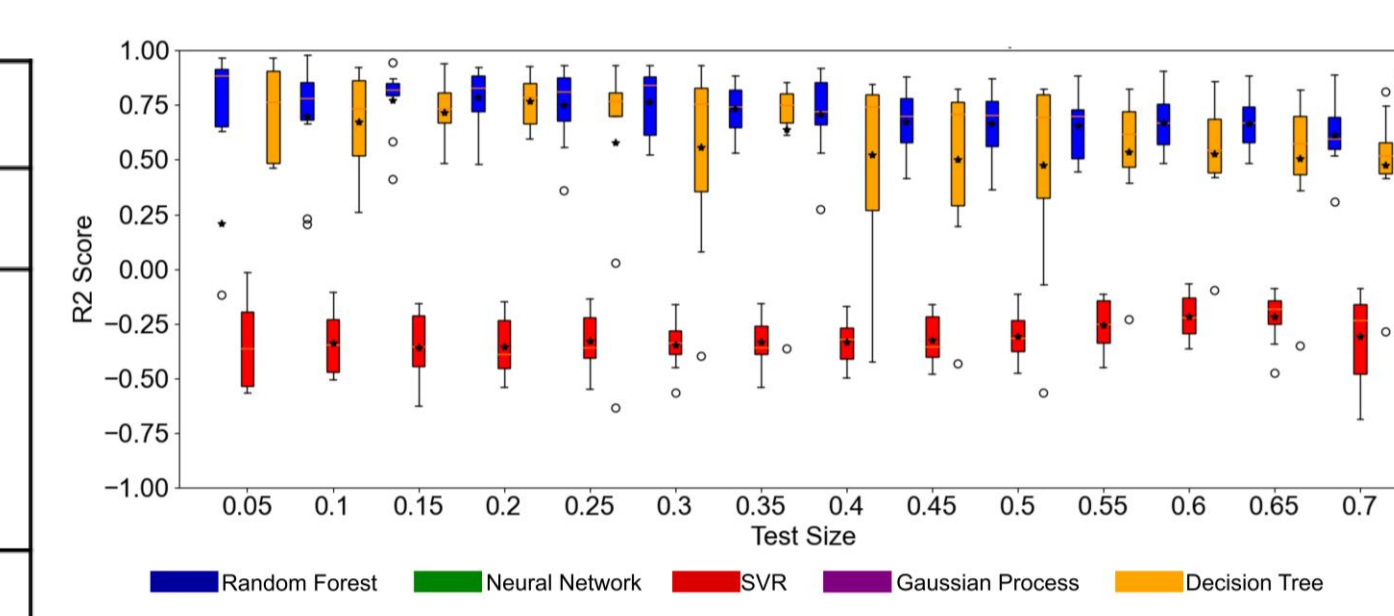


Comparing the observed and recovered variability of different tensile properties

Estimated variability to: $n \sim 1.1\%$; $E \sim 1\%$; $\sigma_y \sim 5\%$

KL Divergence

Group	Property	Model	
		SIP	Bayesian
BY	n	0.0042	0.18
	E	0.0008	0.14
	σ _y	0.0005	0.17
EY	n	0.012	0.25
	E	0.002	0.23
	σ _y	0.003	0.19



Performance of different forward model

SUMMARY

- Presented a stochastic inverse modeling approach for inverse indentation problems
- Applied to additively manufactured stainless steel data for property recovery
- Method shows enhanced accuracy over conventional techniques

FUTURE WORK

- Refine sampling strategy
- Apply method to diverse AM datasets
- Characterize microstructure from ECT data

REFERENCES

- [ASTM, 2023] ASTM International, "Unlocking the Bottleneck" AM CoE, June 2023.
- [Dao et al., 2001] Dao, M., et al. Acta materialia 49.19: 3899-3918 (2001).
- [Kim et al. 2021] Kim, Changgong et al., " Additive ManufacturingVol. 38.